Delta Function :

$$V(x) = -\alpha\delta(x)$$

$$E = \frac{\hbar^2}{mL^2}$$

$$L = \frac{\hbar^2}{m\alpha} \quad , \quad E = \frac{m\alpha^2}{\hbar^2}$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = (E-V)\psi$$

$$\frac{d^2\psi}{dx^2} = \frac{-2mE}{\hbar^2}\psi = k^2\psi$$

Discontinuity

$$\Delta_0\left(\frac{d\psi}{dx}\right) = \lim_{\varepsilon\to 0}\left(\frac{d\psi}{dx}\Big|_{\varepsilon} - \frac{d\psi}{dx}\Big|_{-\varepsilon}\right)$$

$$\boxed{\Delta_0\left(\frac{d\psi}{dx}\right) = \frac{-2m\alpha}{\hbar^2}\psi(0)}$$

$$\Downarrow$$

$$\Delta_0\left(\frac{d\psi}{dx}\right) = -2kA = \frac{-2m\alpha}{\hbar^2}A$$

$$k = \frac{m\alpha}{\hbar^2}$$

$$E = \frac{-\hbar^2 k^2}{2m} = \frac{-1}{2}\frac{m\alpha^2}{\hbar^2}$$

Node Theorem :

Infinite - Square well : $\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$

n-1 nodes

$$E_n = \frac{\hbar^2 n^2}{2ma^2}$$

Harmonic Oscillator :

$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$; with Restoring force $-kx$

$V = \frac{1}{2}kx^2 \quad T = \frac{1}{2}mv^2$

$\omega = \sqrt{\frac{k}{m}}$ → for quadratic $V(x)$

$$E = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

Hamiltonian : $\hat{H} = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 \hat{x}^2$

$$[\hat{x}, \hat{p}] = i\hbar$$

Time - independent Schrodinger eq

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi = E\psi$$

Let $x = au$, $a^2 = \dfrac{\hbar}{m\omega}$

$\dfrac{\hbar^2}{ma^2} = \hbar\omega$, $m\omega^2 a^2 = \hbar\omega$

$$-\frac{1}{2}\hbar\omega \frac{d^2\psi(u)}{du^2} + \frac{1}{2}\hbar\omega u^2 \psi(u) = E\psi(u)$$

Energy Parameter

$$\varepsilon = \frac{2E}{\hbar\omega} \Rightarrow \boxed{E = \frac{1}{2}\hbar\omega\varepsilon}$$

$$\boxed{\frac{d^2\psi(u)}{du^2} = (u^2 - \varepsilon)\psi(u)}$$

dimentionless

$u = ax$

$a = \sqrt{\dfrac{\hbar}{m\omega}}$

a is unit length

$$\Rightarrow \psi(u) = h(u) \cdot e^{-u^2/2}$$

$$h''(u) - 2uh' + (\varepsilon - 1)h = 0$$

: Hermite polynomial.

$H_0(u) = 1$

$H_1(u) = 2u$

$H_2(u) = 4u^2 - 2$

$H_3(u) = 8u^3 - 12u$

$$e^{-z^2 + 2zu} = \sum_{0}^{\infty} \frac{z^n}{n!} H_n(u)$$

$$\psi_n(u) \sim H_n(u) \cdot e^{-u^2/2}$$

$$\psi_n(x) = N_n H_n\left(\sqrt{\frac{m\omega}{\hbar}}\, x\right) e^{-\frac{m\omega}{2\hbar}x^2}$$

$$\boxed{\psi_n(x) = N_n H_n\left(x\sqrt{\frac{m\omega}{\hbar}}\right) e^{\cdots} \qquad n = 0, 1, 2}$$